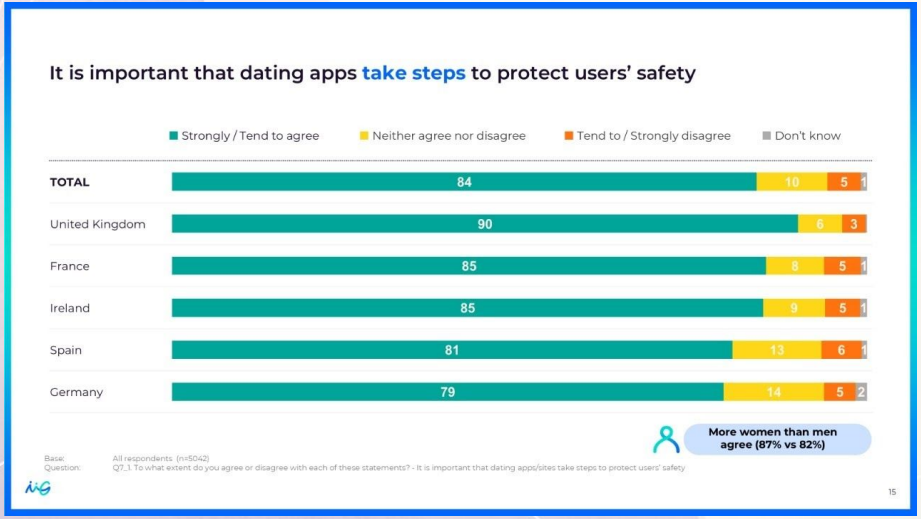
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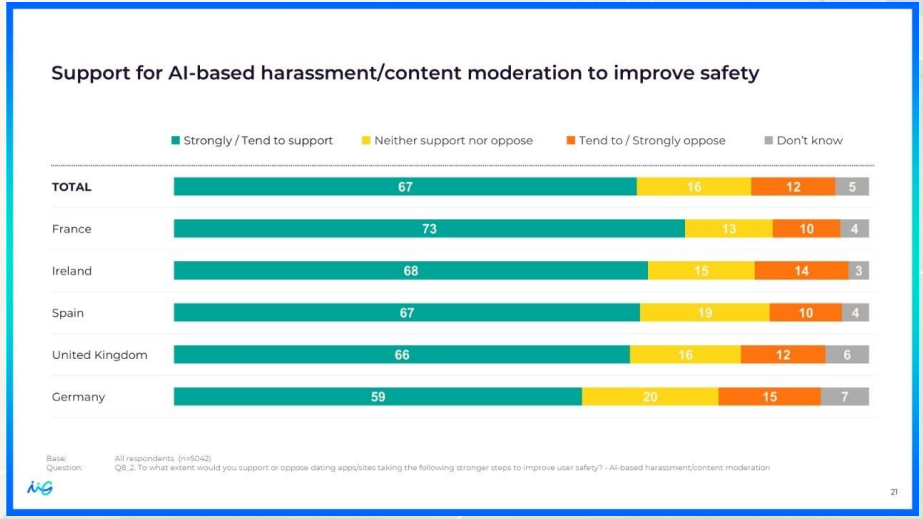
Hybrid Rule + ML + LLM Moderation at Scale

Evolution from raw data to real-time inference

Why moderation matters

Safety is not a support function — it is part of the product experience

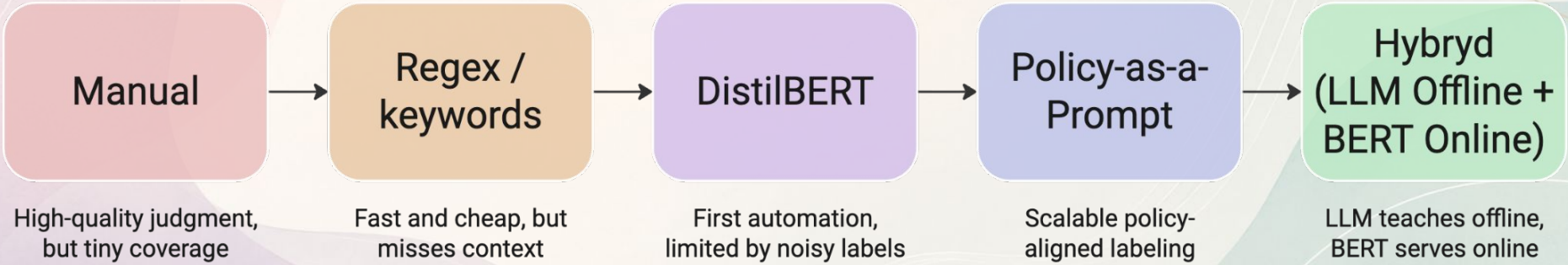




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Evolution timeline

From limited manual coverage to scalable AI moderation: each stage reduced a different bottleneck — coverage, consistency, labeling quality, or inference cost.



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Gray zones

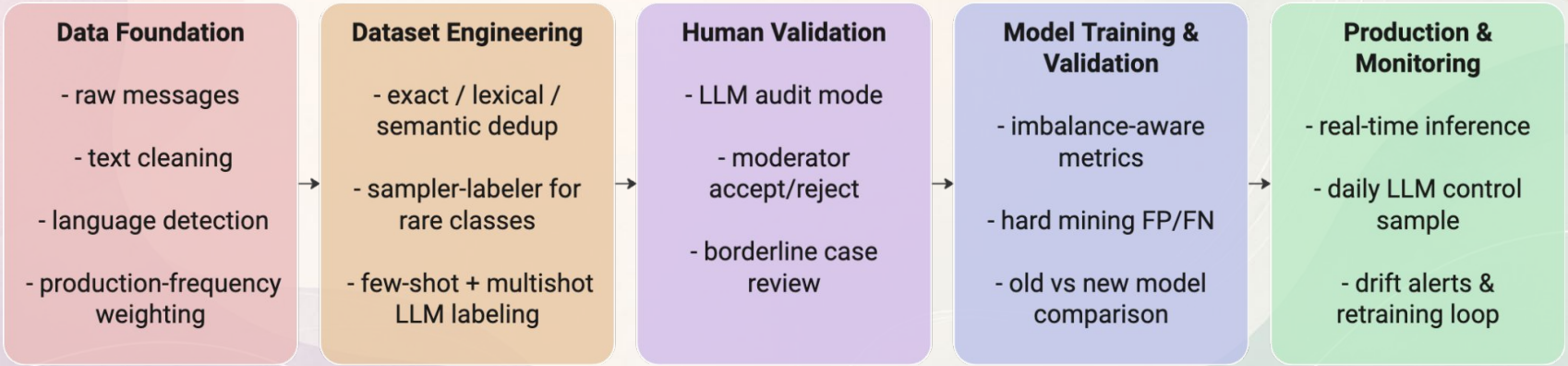
The hardest violations do not look like violations at first glance



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End-to-End Hybrid Moderation Pipeline

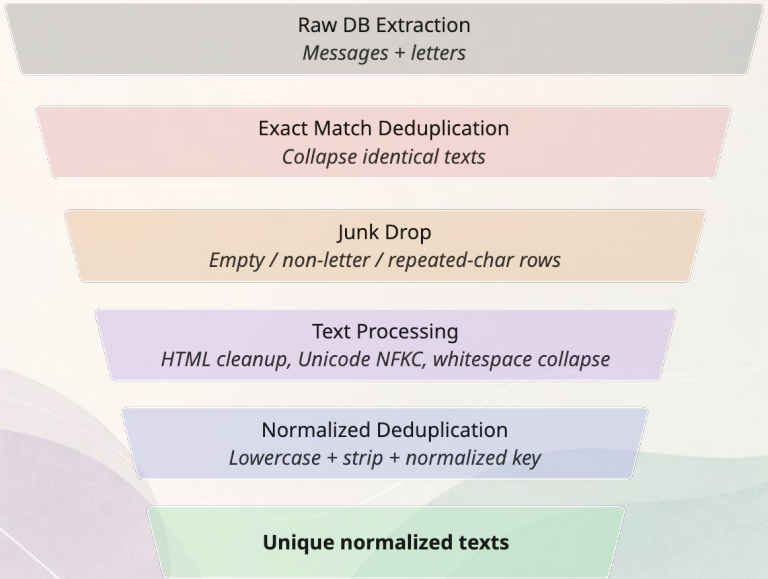
From ambiguous chats to production decisions and feedback loops



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Data Preparation Funnel

Before labeling or training, raw production messages must be cleaned, normalized, and made reproducible.



Clean normalized corpus — not the final training dataset yet

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Adaptive Deduplication: Remove Templates, Preserve Diversity

Start with defaults, inspect clusters, adapt what to keep.

Method	Threshold / Filter	What it catches
1-word-diff	Template hash match	One-name / one-token swaps
Lexical (MinHash)	Jaccard ≥ 0.5	Structurally close templates
Semantic Exact (FAISS)	Cosine ≥ 0.82	Meaning-level paraphrases
Semantic LSH (SimHash)	Cosine ≥ 0.90	Large-scale semantic clusters

Cluster review

For each cluster:
Keep 1 representative
Keep K diverse examples
Keep all if cluster is too diverse

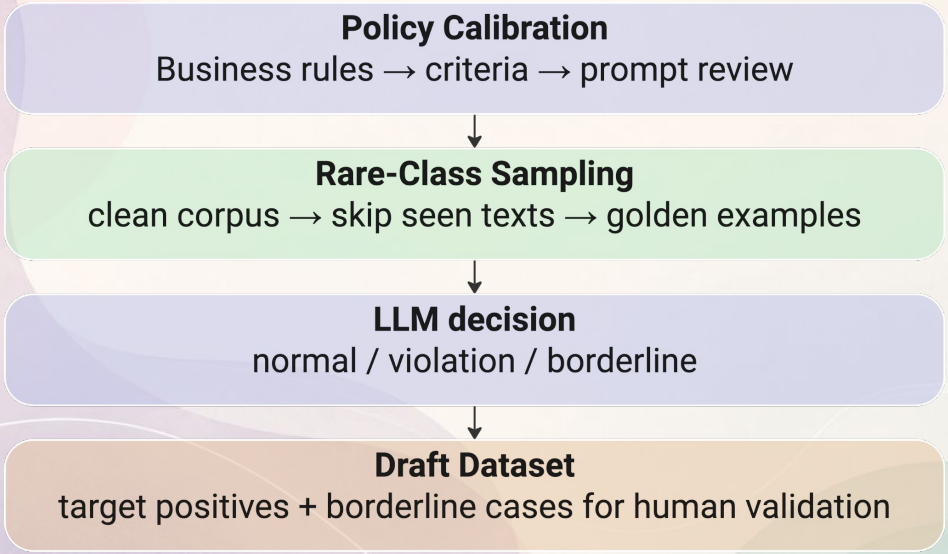
Judge model + human spot-checks

Deduplication is adaptive, not hardcoded.

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Sampler-Labeler: Policy-Aligned Rare-Class Labeling

Business rules are converted into explicit criteria, validated on samples, and then scaled with LLM labeling.



Contrast

Naive labeling

Process 1M texts:
- mostly negatives
-high cost, low signal

Sampler-labeler

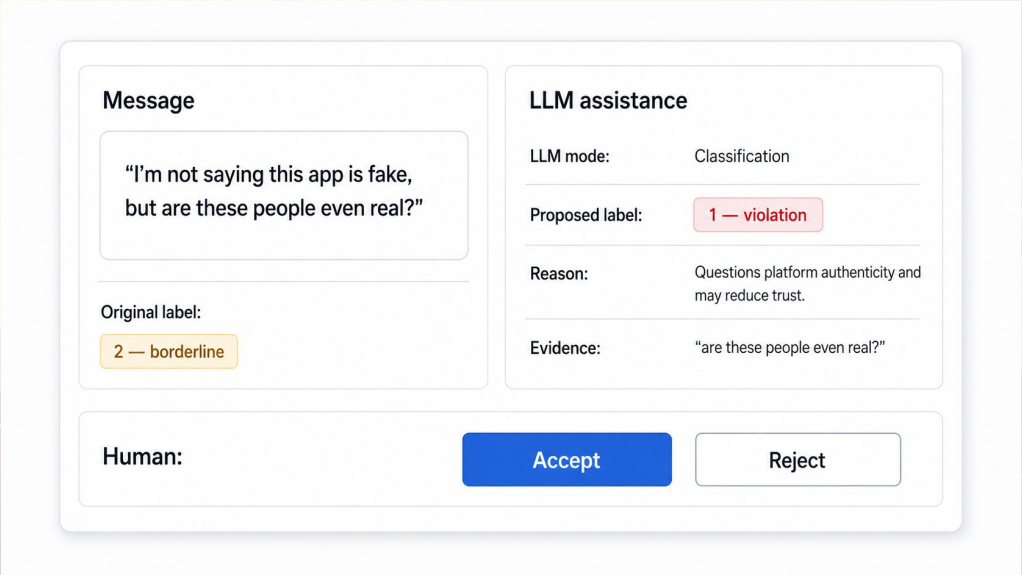
Collect target positives:
- balanced dataset
- better signal per LLM call

We validate the judge on samples before scaling — LLM labels are useful only when the policy is explicit and checked.

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Human-in-the-loop

Raw LLM labels are not ground truth. Without validation, the BERT model would learn LLM mistakes.



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Online Inference Architecture

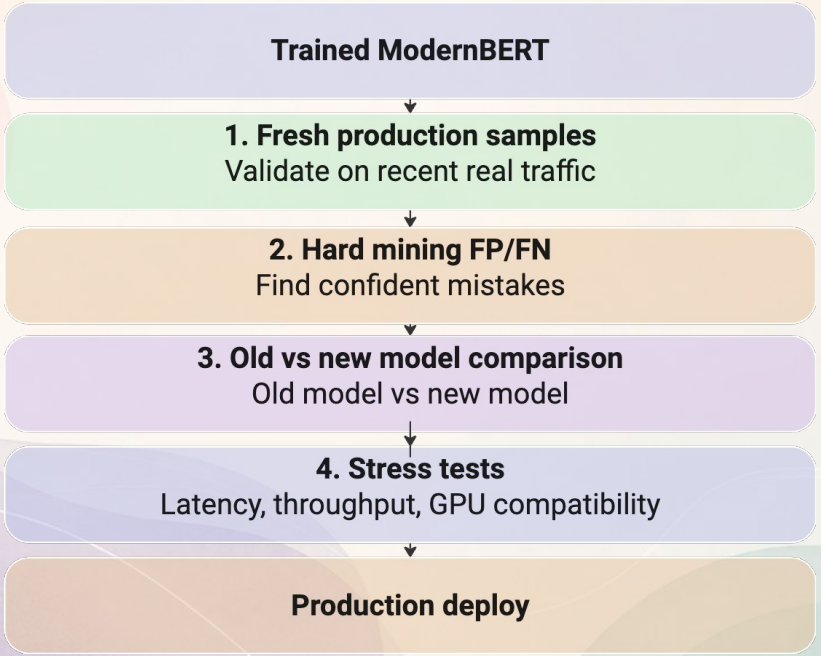
Expensive intelligence offline. Fast decisions online.



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Pre-Deploy Validation: From Training to Production

A model is not ready when it is trained — it is ready when it survives validation



Identity / Parity Check

Same model
Same inputs
Different runtime

Baseline vs optimized
Single vs batch
T4 / L4 / A100

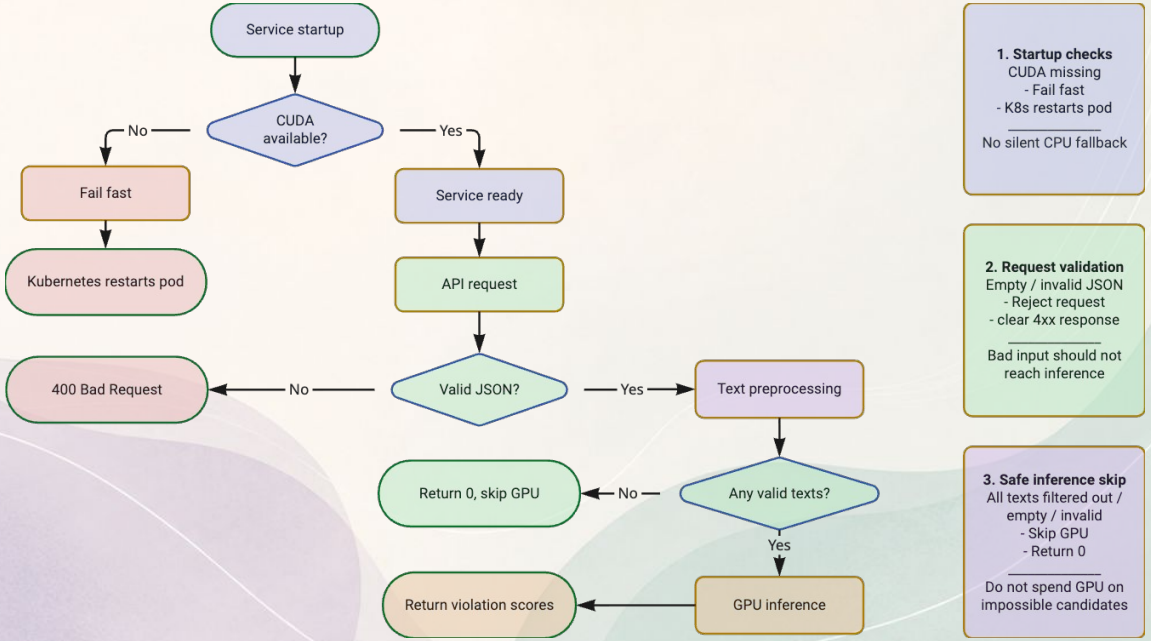
Speed-ups must not change decisions

Speed-ups must not silently change moderation decisions.

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Service Resilience: Fail Fast or Skip Safely

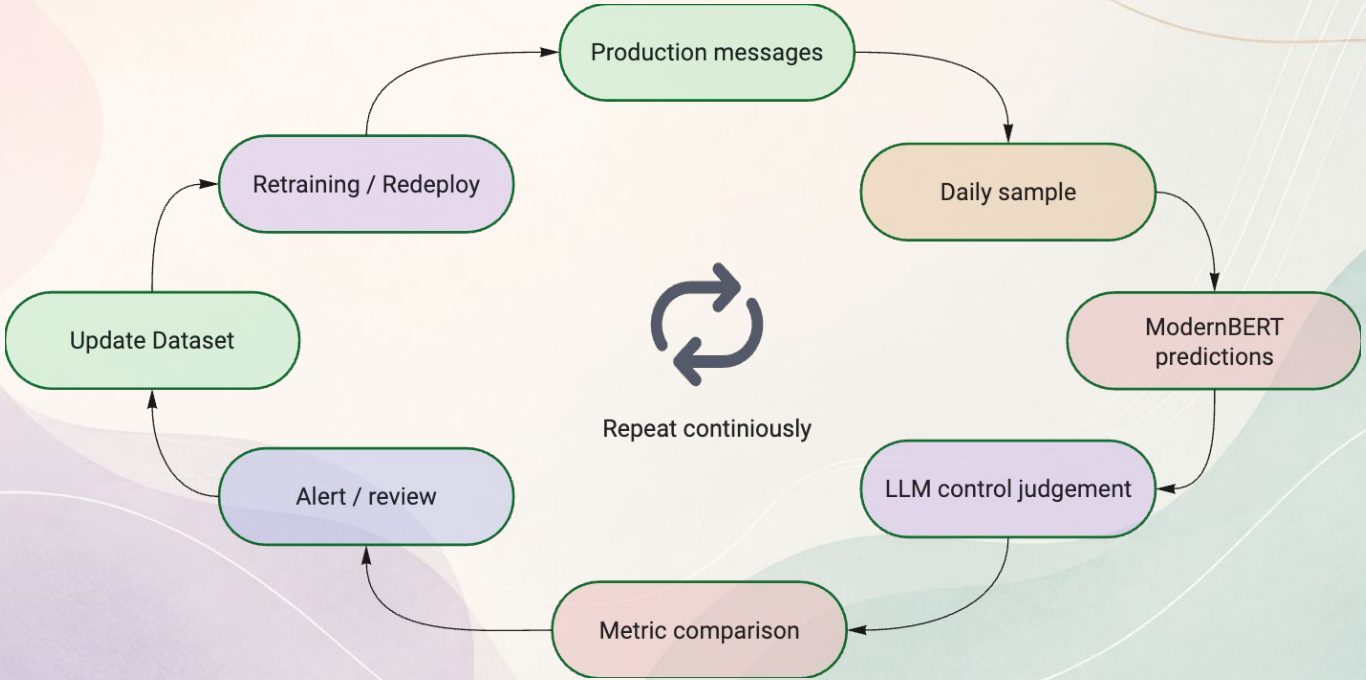
Reliability principle: fail loudly on infrastructure problems, degrade safely on invalid or irrelevant input.



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Drift Monitoring & Feedback Loop

Abuse tactics evolve; monitoring closes the learning loop



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Key Takeaways

Moderation is not a cost center — it is trust infrastructure

1. Moderation is an end-to-end MLOps system
Data cleaning, labeling, deployment, monitoring, retraining.

2. Hybrid architecture gives the best trade-off
LLM labels offline; ModernBERT serves online.

3. Data quality matters as much as model choice
Deduplication, preprocessing, and validation set the ceiling.

4. Production readiness is more than accuracy
Latency, resilience, stress tests, and drift monitoring matter.

The goal is not just to detect violations — it is to protect user trust at scale.

Q&A

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Thank you!

Hybrid Rule + ML + LLM Moderation at Scale

Built with the ML Team at Social Discovery Group

Andrei Shcherbinin
ML Engineer / ML Team Lead
Social Discovery Group

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Discreditation Detection: Finding a Needle in a Haystack

Rare class, high impact: fast filtering + smart verification

